

# HWRS 561b: Physical Hydrogeology II

## Steady-state 1D unsat flow (Part 1)

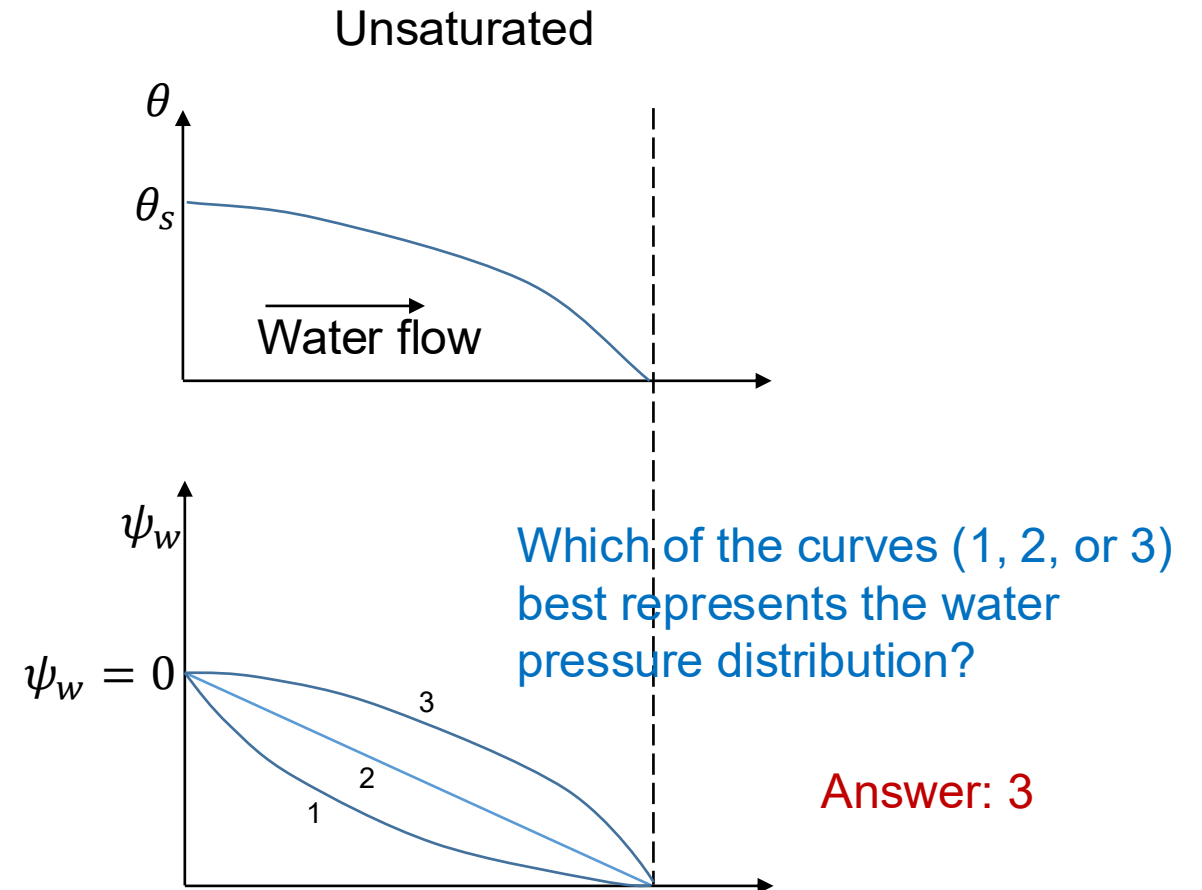
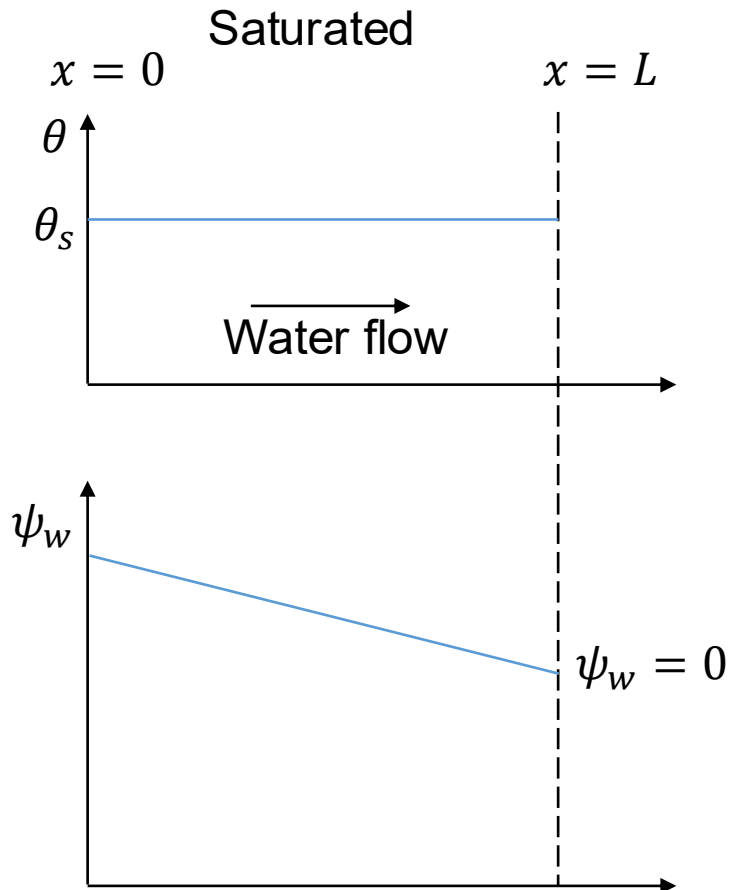
Agenda: Steady-state unsaturated flow  
Reading: Chapter 11 (Pinder & Celia, 2006)

- ❖ Three forms of Richards' equation
  - Mixed form
  - Pressure head-based form
    - ✓ Specific moisture capacity
  - Water content-based form
    - ✓ Soil moisture diffusivity (What does the equation have to do with “diffusion”?)
    - ✓ Cannot be used if the domain involves saturated water flow
  - How to include soil and fluid compressibility?
- ❖ Richards' assumptions
  - Air pressure remains almost zero everywhere, but air does move.
  - Does “air movement” make the Richards' equation invalid? No, as long as air pressure remains almost zero everywhere.

# Steady-State Unsaturated Flow

HWRS 561b  
Bo Guo  
Spring 2026

Horizontal flow (Homogeneous column)

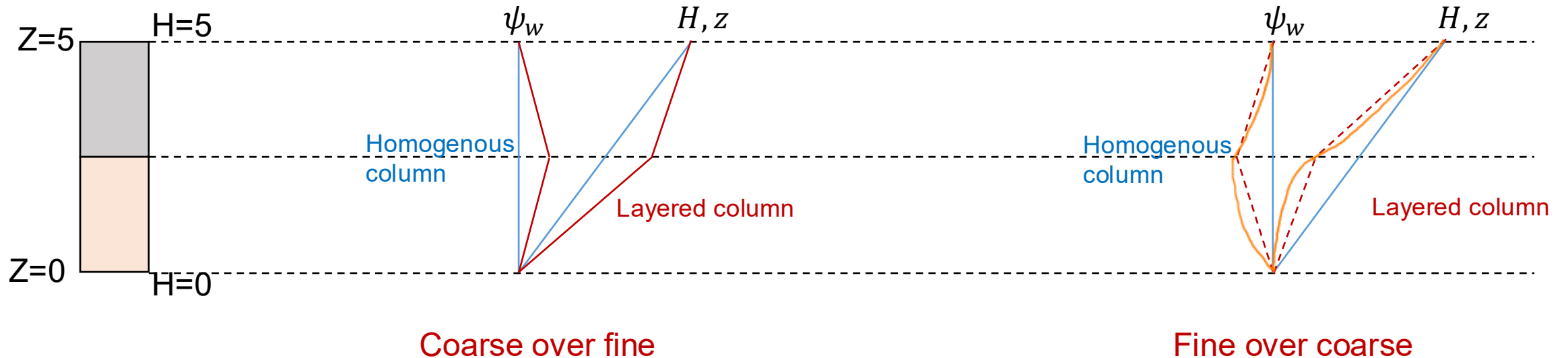


- Unsaturated flow involves nonlinearities that make their behaviors differ from that of the saturated flow

# Steady-State Unsaturated Flow

HWRS 561b  
Bo Guo  
Spring 2026

Vertical flow (in layered columns)

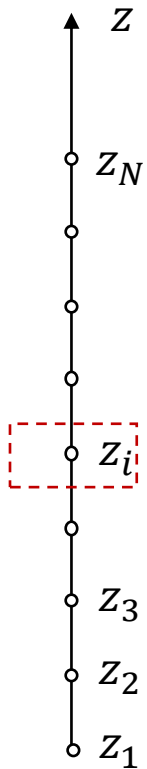


- Hydraulic head and pressure head are both continuous in space.
- Is water saturation continuous?

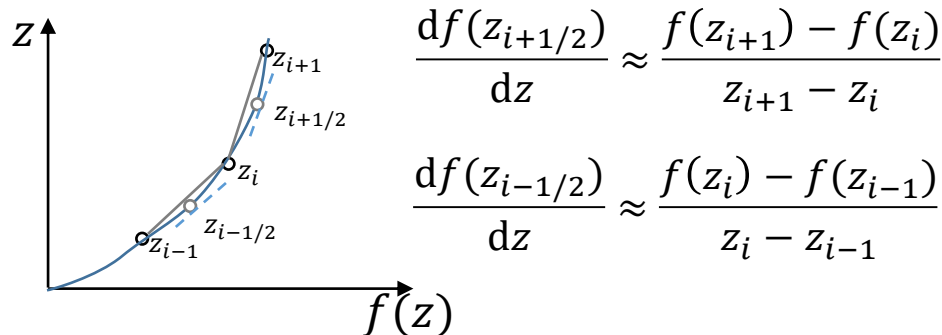
# Steady-State Unsaturated Flow: Numerical Soln.

$$\frac{\partial \theta_w}{\partial t} - \frac{\partial}{\partial z} \left( K \frac{\partial \psi_w}{\partial z} \right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \frac{d}{dz} \left( K \frac{d\psi_w}{dz} \right) + \frac{dK}{dz} = 0 \quad (1)$$

- Equation (1) is a second-order ordinary differential equation in 1D.
- It is nonlinear because  $K = K(\psi_w)$  is a nonlinear function.
- To solve it, we need two boundary conditions and we need to do it in an iterative procedure.

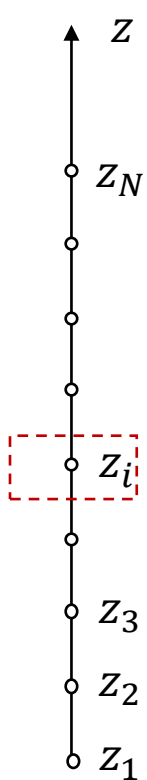


- How to solve this 1D **nonlinear ordinary differential equation**?
- Key idea: Divide the domain into many boxes and convert the **differential equation** to a system of nonlinear **algebraic equations**.
- Technique: Use finite difference to approximate derivatives



# Steady-State Unsaturated Flow: Numerical Soln.

$$\cancel{\frac{\partial \theta_w}{\partial t}} - \frac{\partial}{\partial z} \left( K \frac{\partial \psi_w}{\partial z} \right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \textcircled{1} \frac{d}{dz} \left( K \frac{d\psi_w}{dz} \right) + \textcircled{2} \frac{dK}{dz} = 0 \quad (1)$$



$$\begin{aligned} \textcircled{1} \frac{d}{dz} \left( K \frac{d\psi_w}{dz} \right) \Big|_{z_i} &\approx \frac{\left( K \frac{d\psi_w}{dz} \right)_{i+1/2} - \left( K \frac{d\psi_w}{dz} \right)_{i-1/2}}{\Delta z} \approx \frac{K_{i+1/2} \frac{\psi_{w,i+1} - \psi_{w,i}}{\Delta z} - K_{i-1/2} \frac{\psi_{w,i} - \psi_{w,i-1}}{\Delta z}}{\Delta z} \\ &= \frac{K_{i+1/2}(\psi_{w,i+1} - \psi_{w,i}) - K_{i-1/2}(\psi_{w,i} - \psi_{w,i-1})}{\Delta z^2} \\ \textcircled{2} \frac{dK}{dz} \Big|_{z_i} &\approx \frac{K_{i+1/2} - K_{i-1/2}}{\Delta z} \\ \Rightarrow \frac{K_{i+1/2}}{\Delta z^2} (\psi_{w,i+1} - \psi_{w,i}) - \frac{K_{i-1/2}}{\Delta z^2} (\psi_{w,i} - \psi_{w,i-1}) + K_{i+1/2} \frac{1}{\Delta z} - K_{i-1/2} \frac{1}{\Delta z} &= 0 \\ \Rightarrow \frac{K_{i+1/2}}{\Delta z^2} \psi_{w,i+1} - \left( \frac{K_{i+1/2}}{\Delta z^2} + \frac{K_{i-1/2}}{\Delta z^2} \right) \psi_{w,i} + \frac{K_{i-1/2}}{\Delta z^2} \psi_{w,i-1} + K_{i+1/2} \frac{1}{\Delta z} - K_{i-1/2} \frac{1}{\Delta z} &= 0 \end{aligned}$$

This is an algebraic equation with 3 unknowns. We can write such an algebraic equation for each node or box and we can get  $N$  algebraic equations for the  $N$  unknowns  $(\psi_{w,1}, \psi_{w,2}, \dots, \psi_{w,N})$

# Steady-State Unsaturated Flow: Numerical Soln.

HWRS 561b  
Bo Guo  
Spring 2026

In matrix form:

$$\begin{pmatrix}
 1 & 0 & 0 & & & & \\
 \frac{K_{3/2}}{\Delta z^2} & -\frac{(K_{3/2} + K_{5/2})}{\Delta z^2} & \frac{K_{5/2}}{\Delta z^2} & & & & \\
 & \frac{K_{5/2}}{\Delta z^2} & -\frac{(K_{5/2} + K_{7/2})}{\Delta z^2} & \frac{K_{7/2}}{\Delta z^2} & & & \\
 & & \ddots & \ddots & \ddots & & \\
 & & & \frac{K_{i-1/2}}{\Delta z^2} & -\frac{(K_{i-1/2} + K_{i+1/2})}{\Delta z^2} & \frac{K_{i+1/2}}{\Delta z^2} & \\
 & & & & \ddots & \ddots & \\
 & & & & & \frac{K_{n-1-1/2}}{\Delta z^2} & -\frac{(K_{n-1-1/2} + K_{n-1+1/2})}{\Delta z^2} & \frac{K_{n-1+1/2}}{\Delta z^2} \\
 & & & & & 0 & 0 & 1
 \end{pmatrix}
 \begin{pmatrix}
 \psi_{w,1} \\
 \psi_{w,2} \\
 \psi_{w,3} \\
 \vdots \\
 \psi_{w,i} \\
 \vdots \\
 \psi_{w,n-1} \\
 \psi_{w,n}
 \end{pmatrix}
 =
 \begin{pmatrix}
 B.C. \\
 \frac{K_{3/2} - K_{5/2}}{\Delta z} \\
 \frac{K_{5/2} - K_{7/2}}{\Delta z} \\
 \vdots \\
 \frac{K_{i-1/2} - K_{i+1/2}}{\Delta z} \\
 \vdots \\
 \frac{K_{n-1-1/2} - K_{n-1+1/2}}{\Delta z} \\
 B.C.
 \end{pmatrix}$$

$\Rightarrow \mathbf{F}(\boldsymbol{\psi}_w) \cdot \boldsymbol{\psi}_w = \mathbf{R}$     How to solve this nonlinear system of algebraic equations?  
 $\Rightarrow$  Employ an iterative methods